PHY-112 | Principles of Physics-2

Akiful Islam (AZW)
Spring 2025 | Class #5

Department of Mathematics & Natural Sciences

BRAC UNIVERSITY
Inspiring Excellence

Recap of the previous class!

BRAC UNIVERSITY
Inspiring Excellence

What we studied in Class #4

Refer to Class #4 slides for details!

- What Electric Flux is and what does it measure
- How to measure Electric Flux visually and numerically for uniform electric field
- How does measuring flux for non-uniform field leads to the evaluation of surface integral of $\vec{E}$.
- How does Gauss's Law simplify to Coulomb's law for point charges?

What is our intention with Gauss's Law?

BRAC UNIVERSITY
Inspiring Excellence

Gauss's Law and Electric fields

How much field through an area?

The total Φ_E passing through a closed surface is proportional to the total electric charge Q_{enc} enclosed within that surface.

$$\oint_S \Phi_E = \oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enclosed}}}{\epsilon_0}. \qquad \text{(Integral Form)}$$

$$\text{where} \quad \epsilon_0 = 8.854 \times 10^{-12}\,\text{F}\,\text{m}^{-1}$$

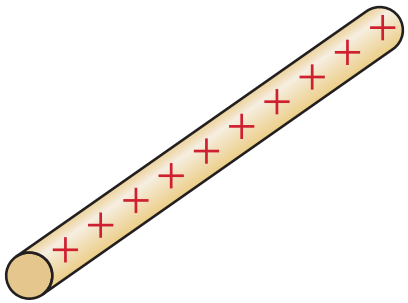
$$\rightarrow \text{Permittivity of free space}$$

Note: Electric Flux Is Independent of Surface Shape and Radius.

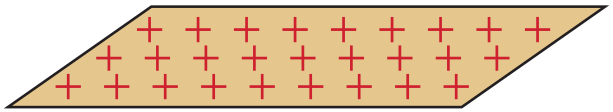
3 Key $\vec{E}$ Field Sources that we will study

Keep the intensity the same all over

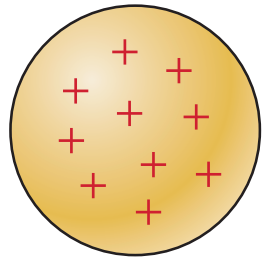
Line Charge



Surface Charge



Volume Charge



The template we shall follow
To simplify our field calculation. For now!!

$$\oint_S \Phi_E = \oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\oint E da \cos\phi = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\pm E \oint da = \frac{Q_{\text{enc}}}{\epsilon_0}; \text{ provided } \cos\phi = \pm 1$$

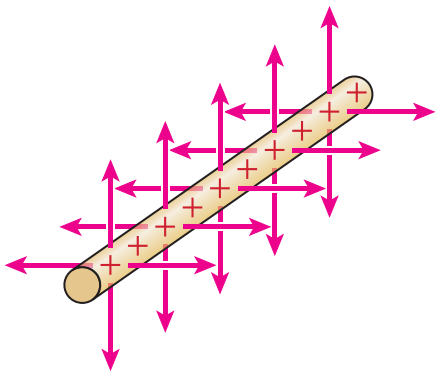
$$EA = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E = \frac{1}{A} \times \frac{Q_{\text{enc}}}{\epsilon_0};$$

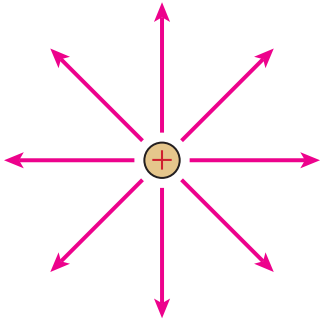
where $Q_{\text{enc}} = \int dq_{\text{enc}} = \text{charge density} \times \text{distribution element}$

$\vec{E}$ Field for an Infinitely long Line Charge

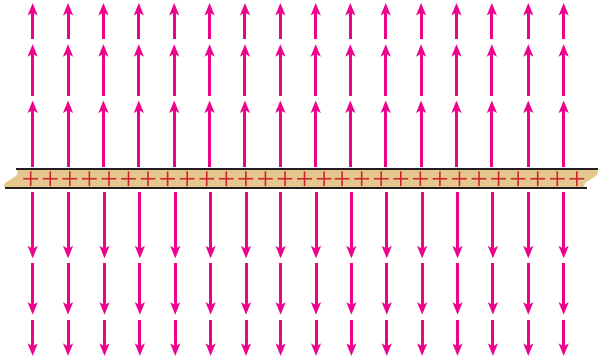
Charge is uniformly distributed across a line



3D view

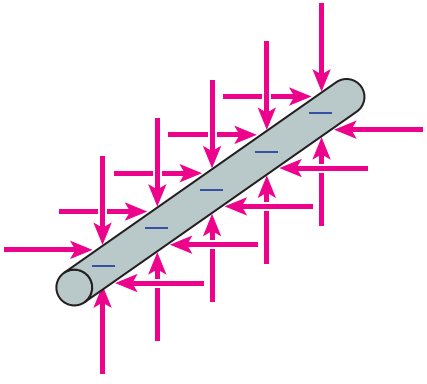


End view

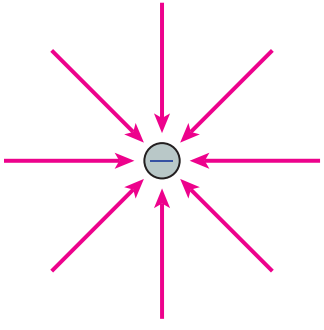


Side view

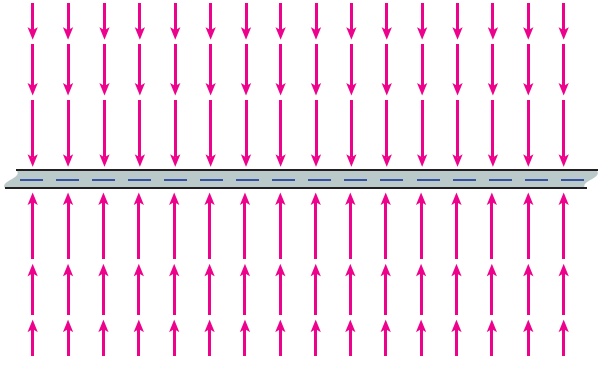
$$\lambda = \frac{Q}{L}$$



3D view



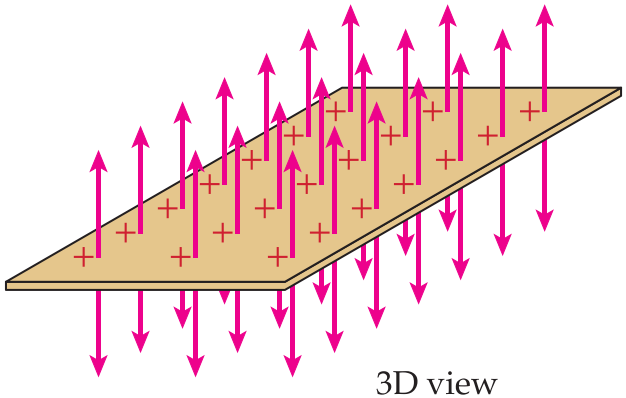
End view



Side view

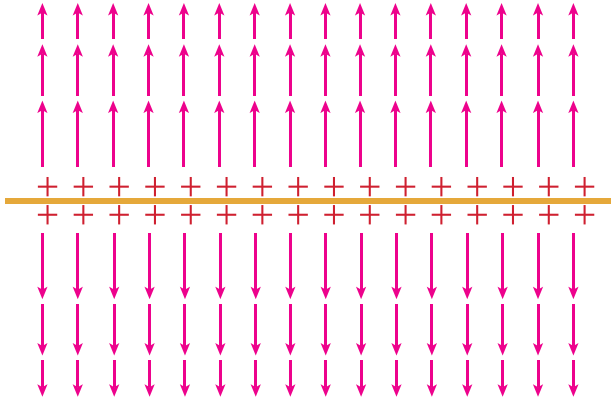
$\vec{E}$ Field for an Infinitely distributed Surface Charge

Charge is uniformly distributed across a surface

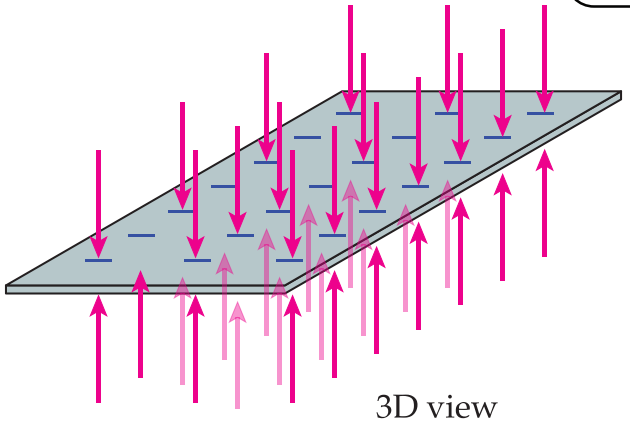


3D view

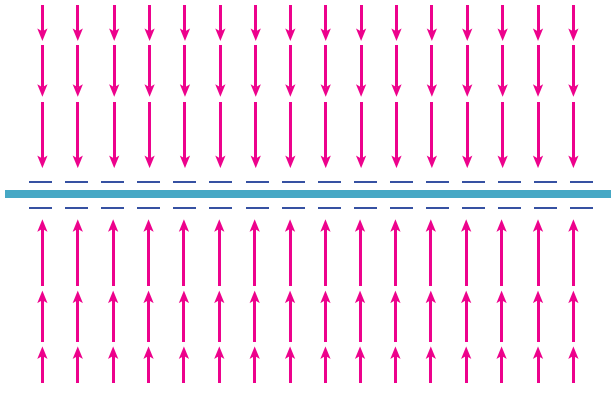
$$\sigma = \frac{Q}{A}$$



End view = Side view



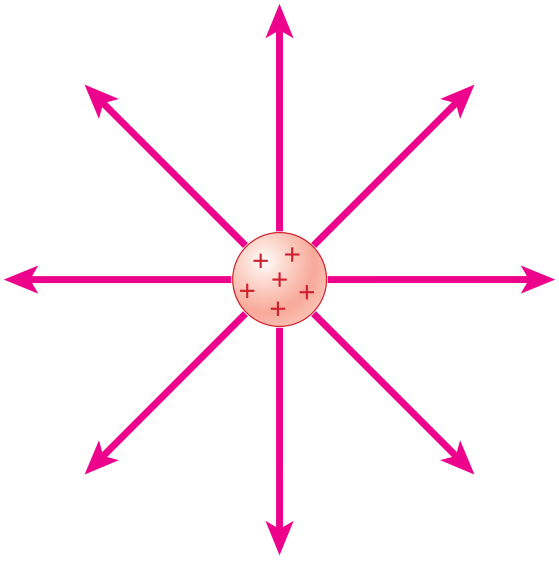
3D view



End view = Side view

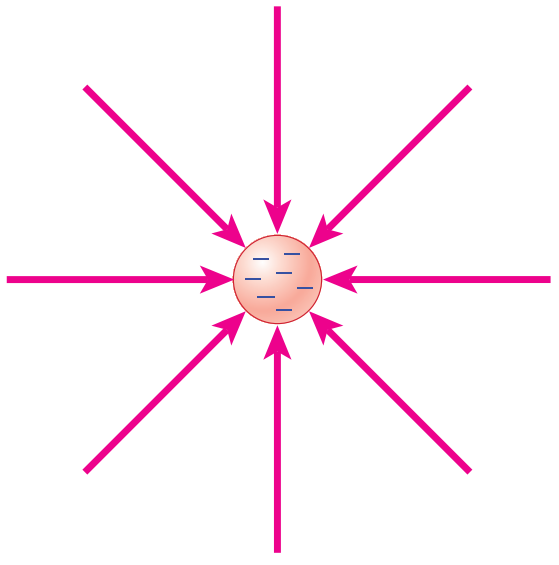
$\vec{E}$ Field for a Volume Charge

Charge is uniformly distributed across a volume



2D view

$$\rho = \frac{Q}{V}$$



2D view

Field acts radially outward (positive), or inward (negative)

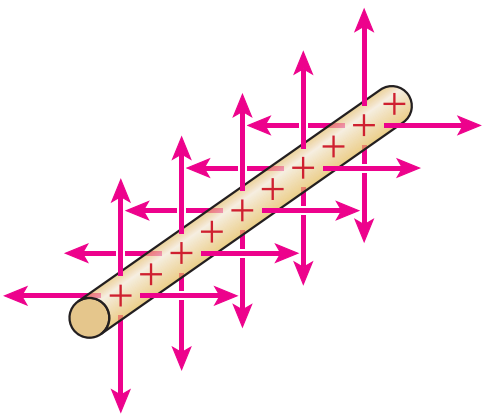
Suitable Gaussian Surfaces

BRAC UNIVERSITY
Inspiring Excellence

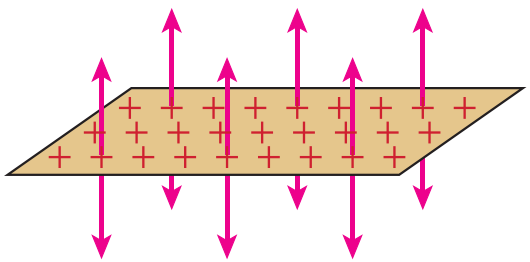
Suitable Gaussian Surfaces for 3 Key $\vec{E}$ Field Sources

Keep the intensity the same all over

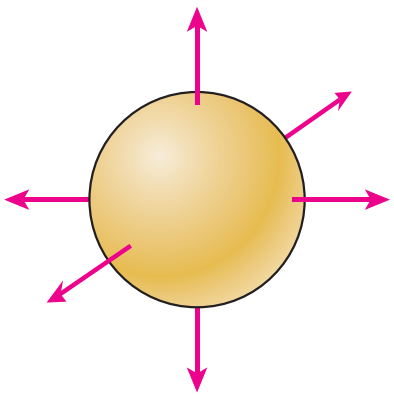
Cylindrical symmetry



Planar symmetry



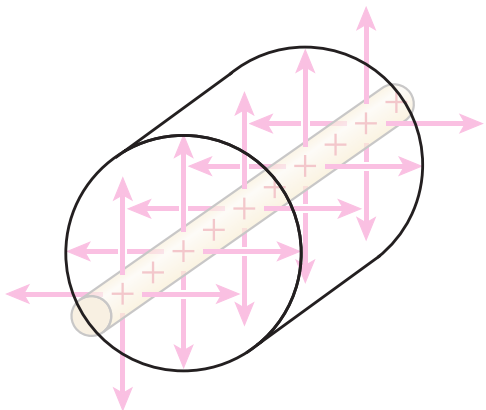
Spherical symmetry



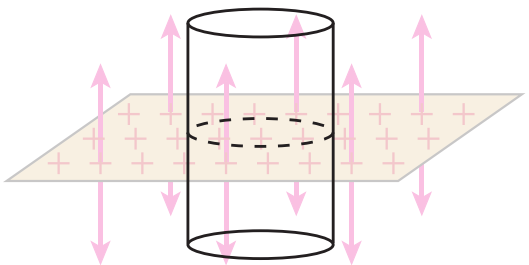
Suitable Gaussian Surfaces for 3 Key $\vec{E}$ Field Sources

Keep the intensity the same all over

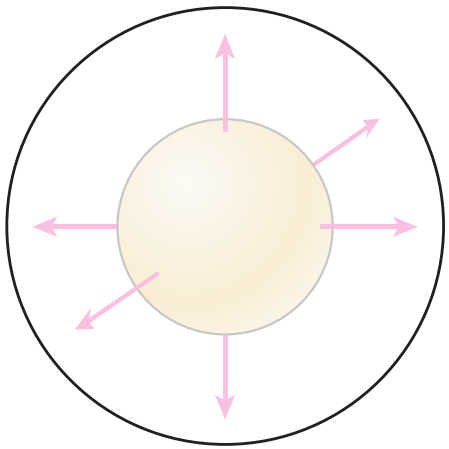
Cylindrical symmetry



Planar symmetry



Spherical symmetry



Suitable Gaussian Surfaces for 3 Key $\vec{E}$ Field Sources

Glance over at one go

- Line Charge $\longrightarrow$ Current Wire $\longrightarrow$ Long Cylindrical
- Surface Charge $\longrightarrow$ Capacitors $\longrightarrow$ Wide Planar
- Volume Charge $\longrightarrow$ Electrodes/Shell Charges $\longrightarrow$ Spherical

Now shut up! and Calculate

BRAC UNIVERSITY
Inspiring Excellence

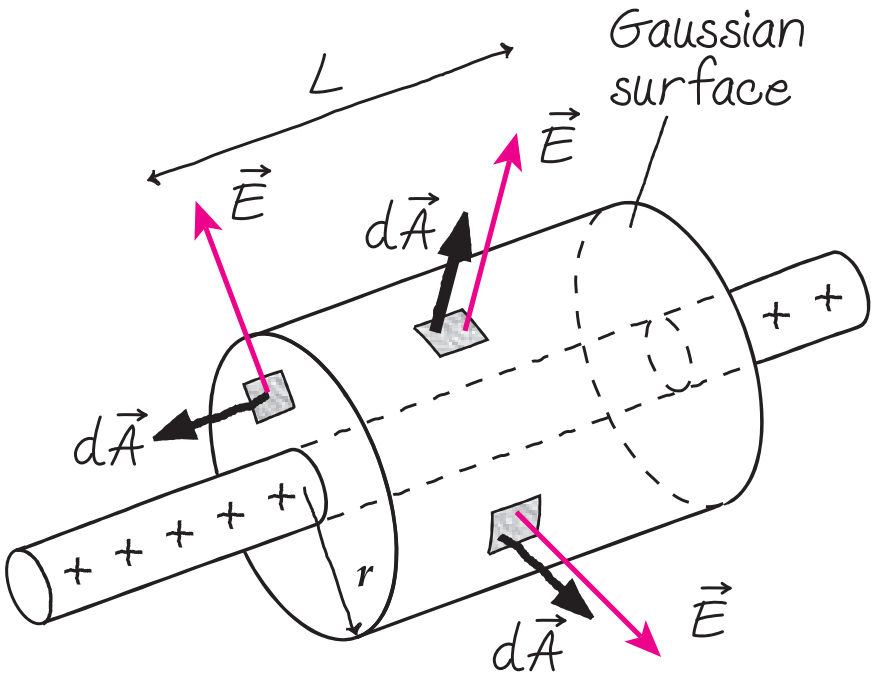
How to apply Gauss's Law to measure $\vec{E}$ fields

One size fits all recipe for easier calculation

1. Check the source and figure out the symmetry of its $\vec{E}$ fields
2. Choose a Gaussian surface matching the field symmetry
3. Calculate the Q_{enc} enclosed by the chosen surface
4. Collect only the area portion of the chosen surface that can record flux, discard the rest
5. Confine everything to one side, keeping E on the left after You have applied Gauss's law. Et voilà!!

$\vec{E}$ Field for an Infinitely long Line Charge: Positive

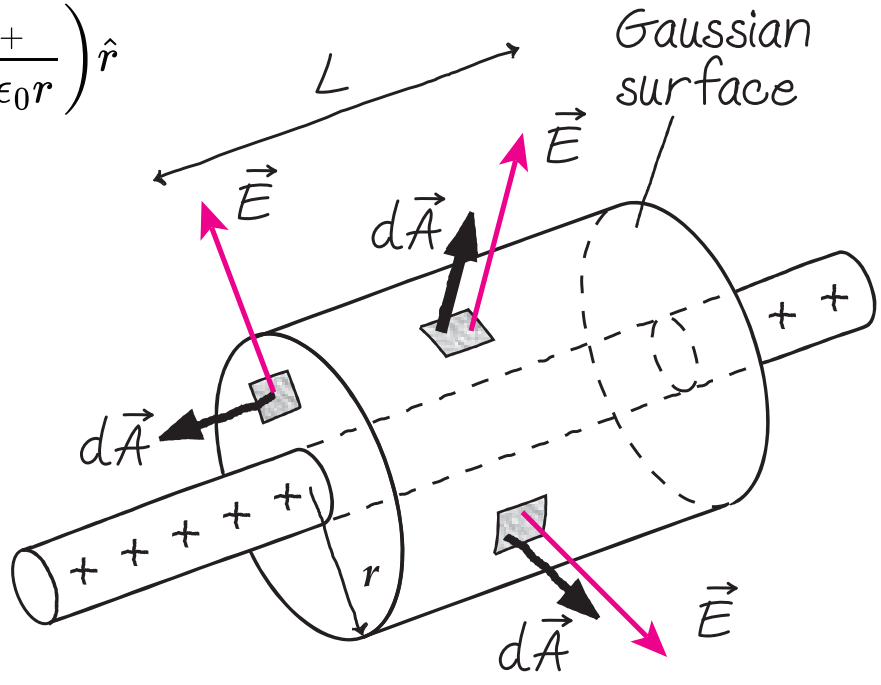
Positive Charge is distributed uniformly over a surface



$\vec{E}$ Field for an Infinitely long Line Charge: Positive

Positive Charge is distributed uniformly over a surface

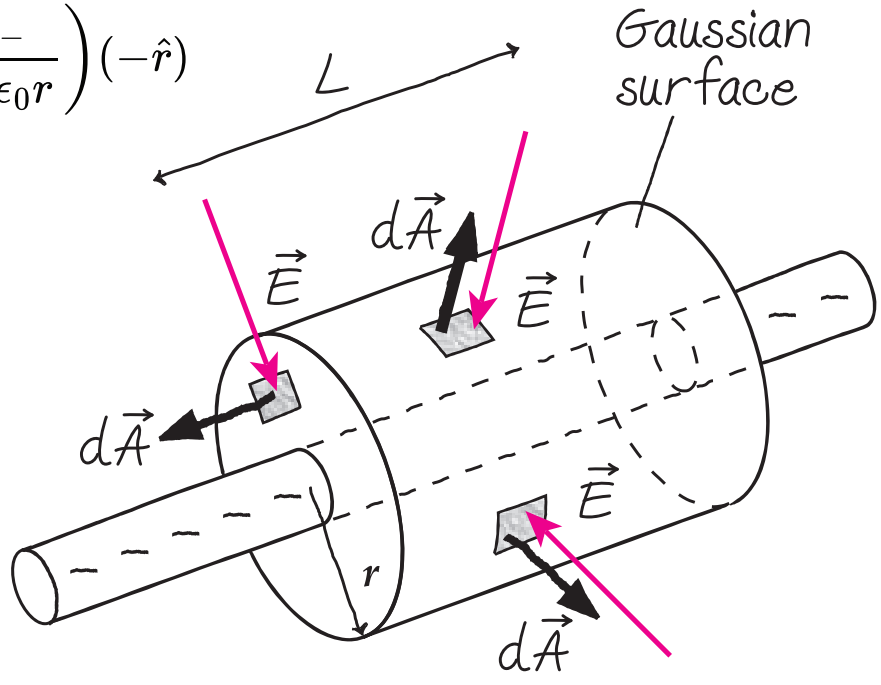
$$\vec{E} = \left(\frac{\lambda_+}{2\pi\epsilon_0 r}\right)\hat{r}$$



$\vec{E}$ Field for an Infinitely long Line Charge: Negative

Negative Charge is distributed uniformly over a surface

$$\vec{E} = \left(\frac{\lambda_-}{2\pi\epsilon_0 r}\right)(-\hat{r})$$



Incepting Ideas (1)

Hint: λ is fixed

Q1: A very long uniform line of charge has a charge per unit length 4.80 µC and lies along the x-axis. A second long uniform line of charge has charge per unit length -2.40 µC and is parallel to the x-axis at $y = 0.400$ m. What is the net electric field (magnitude and direction) at the following points on the y-axis: (a) $y = 0.200$ m and (b) $y = 0.800$ m? (Exercise 22.17, p-770 | Young-Freedman)

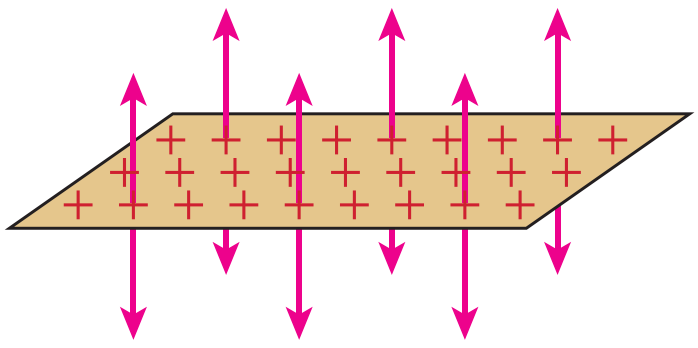
Q2: The electric field 0.335 m from a very long uniform line of charge is 840 N C^{-1}. How much charge is contained in a section of the line of length 2.80 cm? (Exercise 22.18, p-770 | Young-Freedman)

Enter Surface Charge

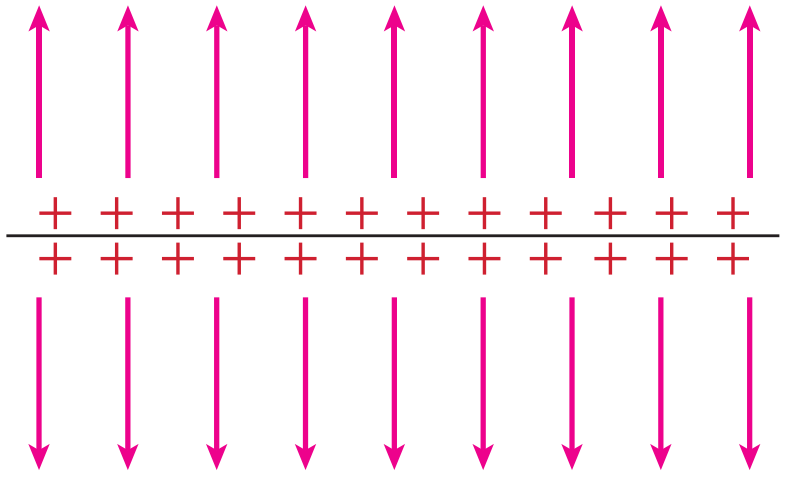
BRAC UNIVERSITY
Inspiring Excellence

$\vec{E}$ Field for a Sheet Charge

Charge is distributed uniformly over a surface



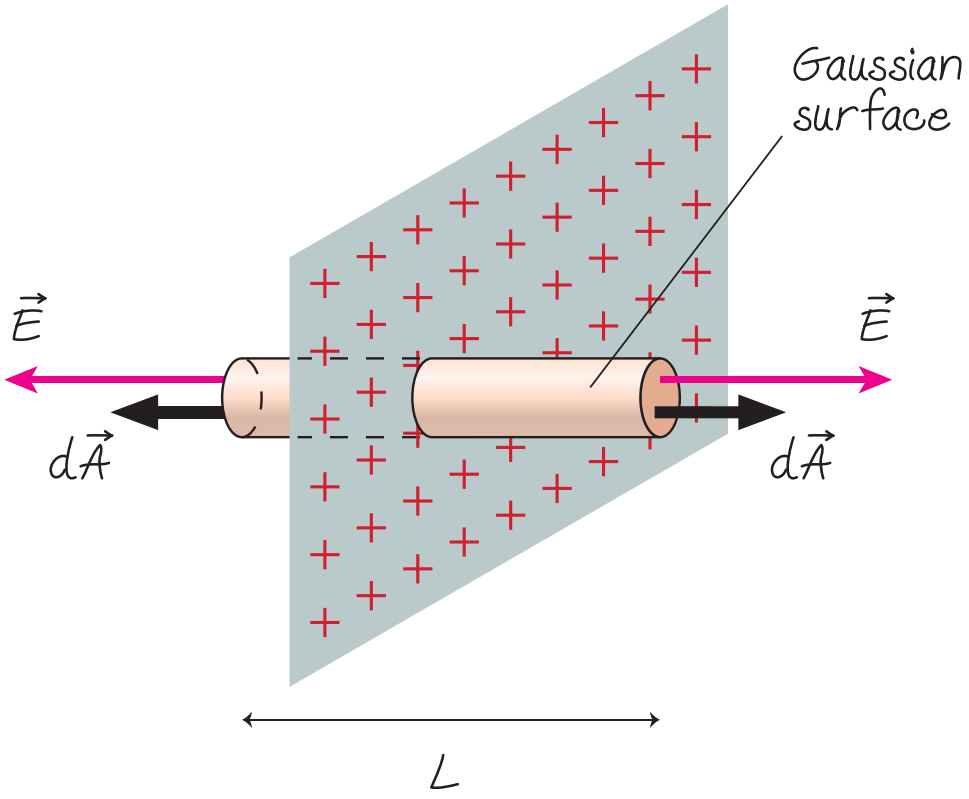
3D view



Side view

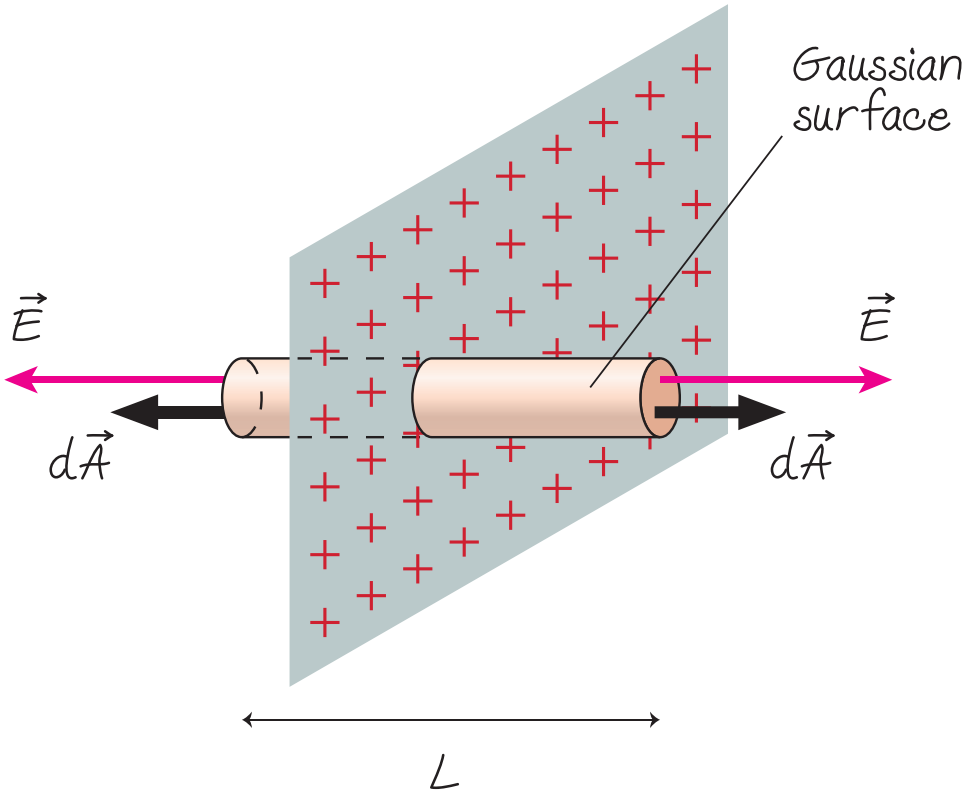
$\vec{E}$ Field for a Sheet Charge

Charge is distributed uniformly over a surface



$\vec{E}$ Field for a Sheet Charge: Positive

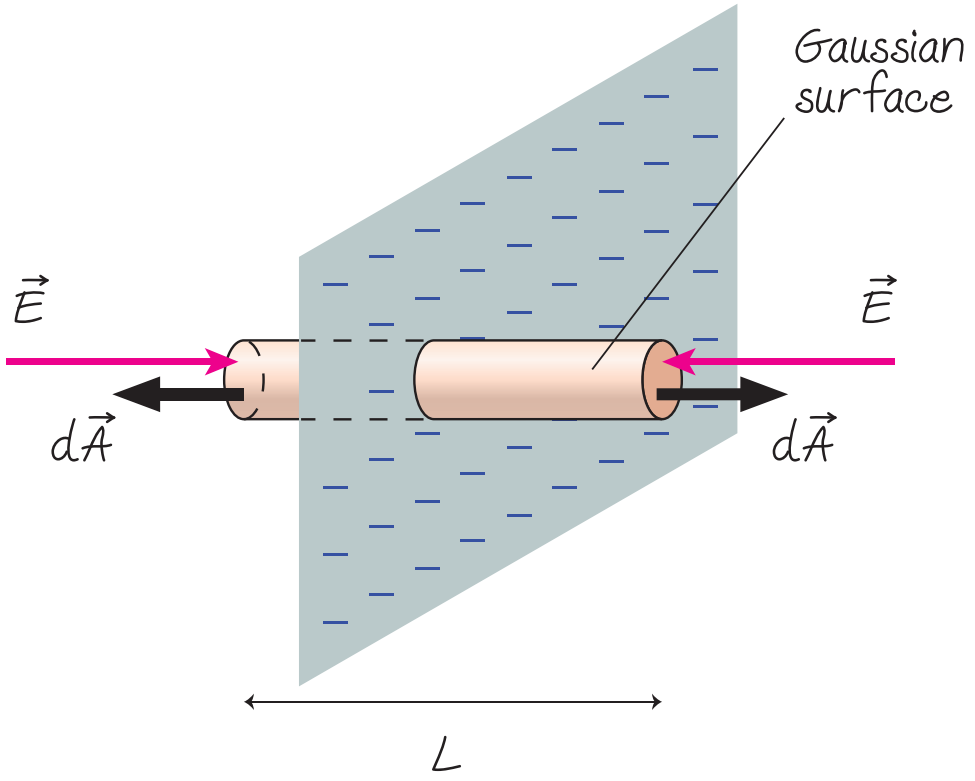
Positive Charge is distributed uniformly over a surface



$$\vec{E} = \left(\frac{\sigma_+}{2\epsilon_0}\right)\hat{n}$$

$\vec{E}$ Field for a Sheet Charge: Negative

Negative Charge is distributed uniformly over a surface

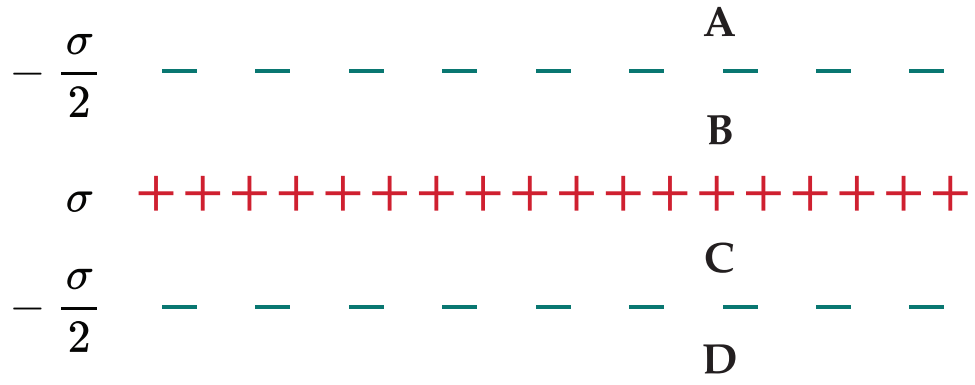


$$\vec{E} = \left(\frac{\sigma_-}{2\epsilon_0}\right)(-\hat{n})$$

Incepting Ideas (2)

Hint: Return of the Superposition Principle

Q: The three parallel planes of charge shown, having surface charge densities $-0.5\sigma, \sigma, -0.5\sigma$. Find the electric fields $\vec{E}_A$ to $\vec{E}_D$ in regions A to D. The upward direction is the $+y$-direction.

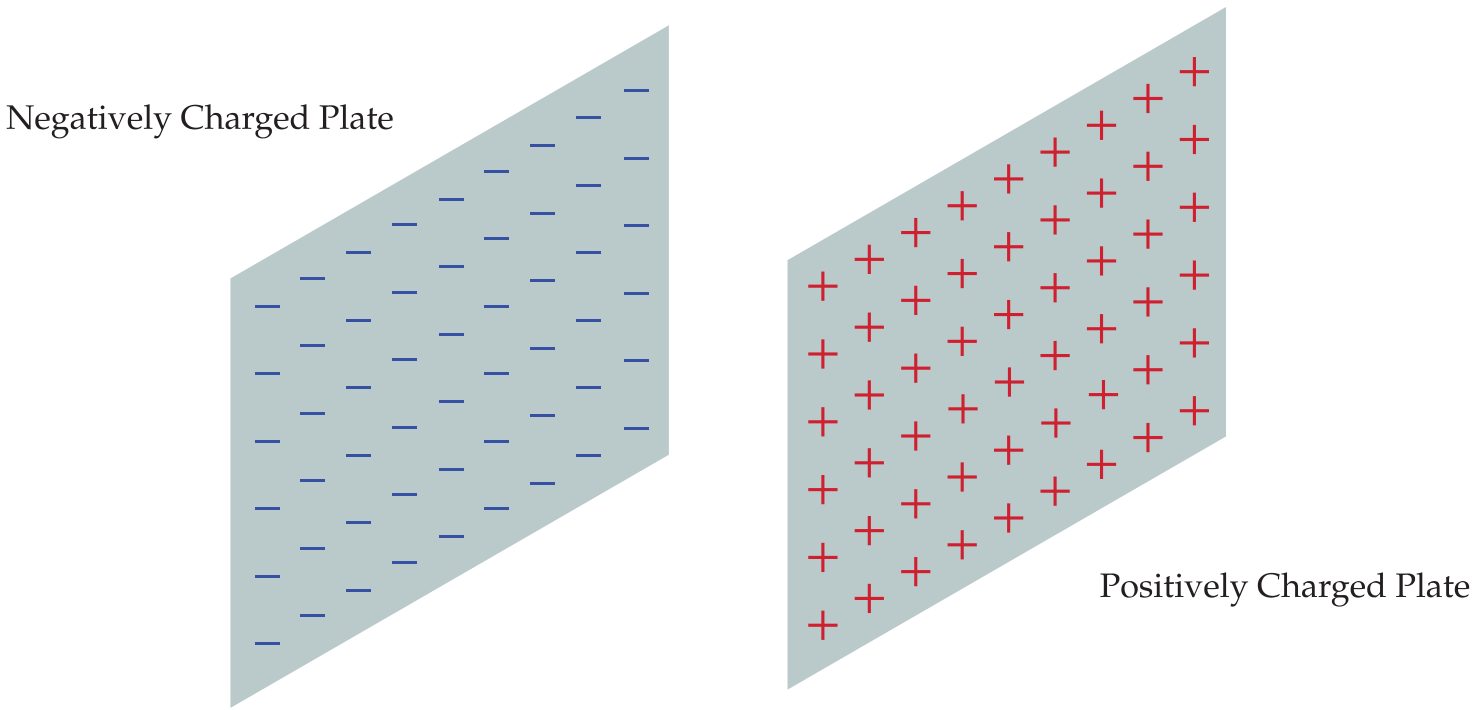


Similar problem to look at for Additional Practice at home

Exercises 22.28, p-770 | Young-Freedman
Sample Problem 23.07, p-674 | Resnick-Halliday
Exercise Problem 69, p-684 | Resnick-Halliday

$\vec{E}$ Field for a Pair of Oppositely Charged Sheet

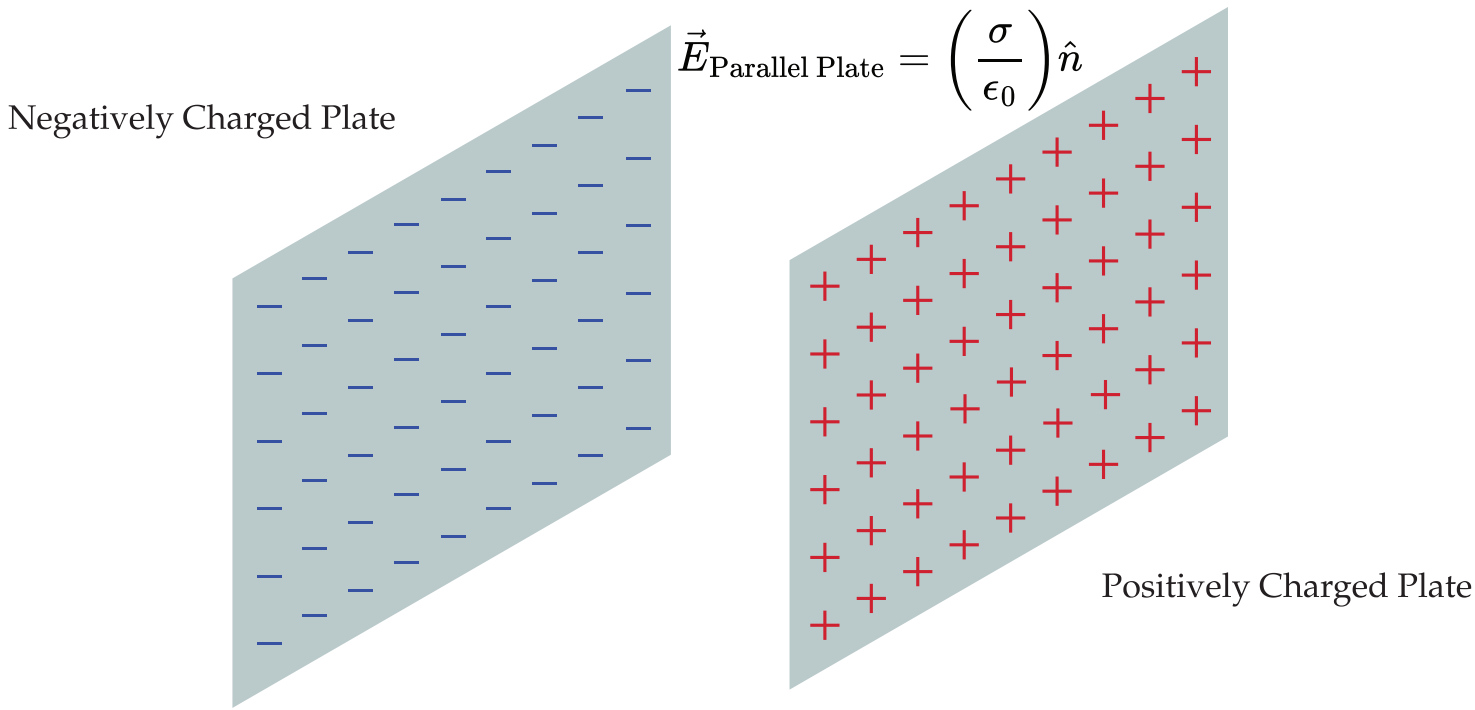
Charge is distributed uniformly over each surface



$\vec{E}$ Field for a Pair of Oppositely Charged Sheet

Charge is distributed uniformly over each surface

$$\vec{E}_{\text{Parallel Plate}} = \left(\frac{\sigma}{\epsilon_0}\right)\hat{n}$$



Incepting Ideas (3)

Hint: Return of the Superposition Principle

Homework Practice Problem: Try it Yourself

Q: Two large, thin metal plates are parallel and close to each other. On their inner faces, the plates have excess surface charge densities of opposite signs and magnitude $7.00 \times 10^{-22}\,\mathrm{C\,m^{-2}}$. In unit-vector notation, what is the electric field at points (i) to the left of the plates, (ii) to the right of them, and (iii) between them?

Similar problem to look at for Additional Practice at home

Exercises 22.28, p-770 | Young-Freedman
Sample Problem 23.07, p-674 | Resnick-Halliday

Incepting Ideas (4)

Practice! Practice! Practice!

Some Practice Problem to try on today's topics

Example Problem 22.7, p-759, 22.8, p-760 | Young-Freedman
Exercise Problem 22.13, 22.17, 22.23, 22.26, p-770 | YF
Checkpoint 3, p-666, 4, p-677 | Resnick-Halliday
Sample Problem 23.07, p-674 | RH
Exercise Problem 4, 5, 6, p-678, 35, p-681, 40, p-682 | RH

That is it for today!

BRAC
Inspiring Excellence